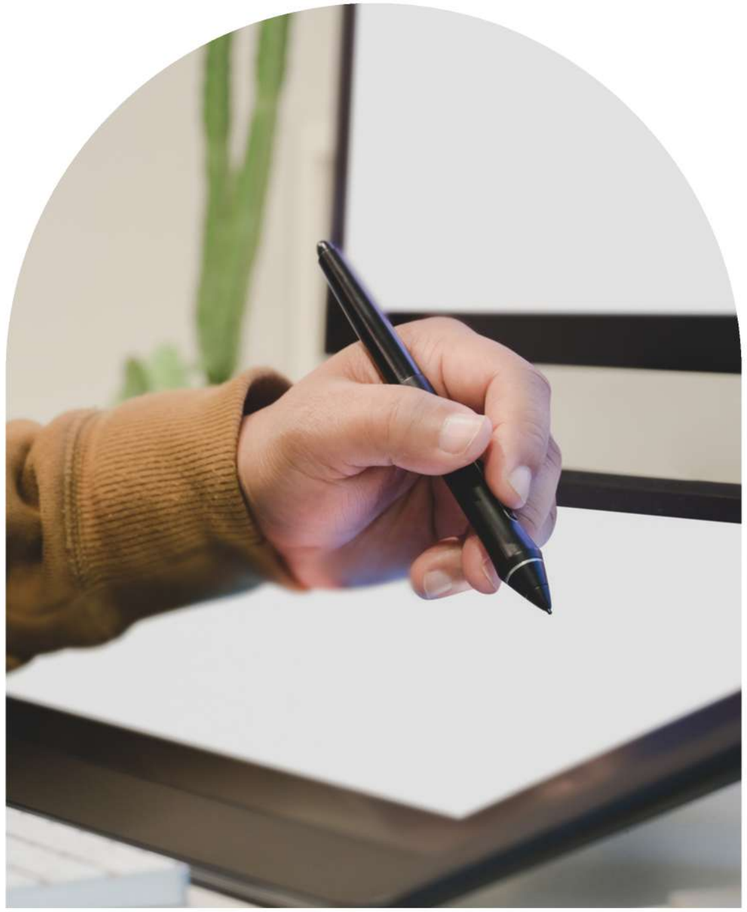
RESUME CLASSIFICATION AND QUALITY DETECTION USING MACHINE LEARNING

Introduction

Machine Learning is a part of Artificial Intelligence that allows computers to learn from data and make decisions. In recruitment, screening resumes manually is time-consuming and inefficient.

This project focuses on building a Resume Classification System that automatically analyzes resumes and predicts their quality. The system uses TF-IDF (Term Frequency-Inverse Document Frequency) to convert text into numerical form and applies Logistic Regression for classification.

This helps in reducing manual effort and improves the efficiency of the hiring process.



Project Title & Problem Statement

Title:

Resume Classification and Quality Detection using Machine Learning

Problem Statement:

Manual resume screening is time-consuming and inefficient. The problem is to automatically classify resumes and detect their quality using machine learning techniques to improve recruitment efficiency.

Explanation of the Project (System Overview)

This system takes raw resume text as input and outputs a predicted label (e.g., Selected/Rejected or Relevant/Not Relevant).

Pipeline:

1. Data Collection (resume dataset)
2. Text Preprocessing (cleaning, tokenization, stopword removal)
3. Feature Extraction (TF-IDF vectors)
4. Model Training (Logistic Regression)
5. Prediction & Evaluation

Workflow:

Resume Text → Cleaning → TF-IDF → ML Model → Prediction → Evaluation

DESCRIPTION OF MODEL / ALGORITHMS USED

Text Preprocessing

- Lowercasing
- Removing special characters (regex)
- Stopword removal (NLTK)
- Tokenization

Feature Extraction

- TF-IDF (Term Frequency–Inverse Document Frequency)Converts text into numerical vectors
- Gives higher importance to meaningful words (e.g., “Python”, “Machine Learning”)

Classification Model

- Logistic RegressionSupervised learning algorithm
- Used for binary classification
- Predicts probability using sigmoid function:
- P(y=1 ∣ x)=11+e−zP(y=1|x) = \frac{1}{1 + e^{-z}}P(y=1 ∣ x)=1+e−z1

Why Logistic Regression?

- Fast and efficient
- Works well for text classification
- Interpretable coefficients

Accuracy / Performance Metrics

1. Accuracy

- Accuracy measures the overall correctness of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- TN (True Negative) → Correctly predicted rejected resumes
- Shows how many resumes are classified correctly out of total resumes.
- Limitation:
- If data is imbalanced, accuracy alone is not reliable.

2. Precision

- Precision measures how many predicted positive results are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

- Precision=TPTP+FPPrecision = \frac{TP}{TP + FP}Precision=TP+FPTP Important when false positives must be minimized
- Out of resumes predicted as “Selected”, how many are actually good resumes.

3.Recall (Sensitivity)

- Definition:
- Recall measures how many actual positives are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

- Important when missing good resumes is risky‘
- Out of all good resumes, how many the model successfully identified.

4.F1-Score

- Definition:
- F1-score is the harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Balances precision and recall
- Useful when data is imbalanced
- Gives a single performance score combining both correctness and completeness.

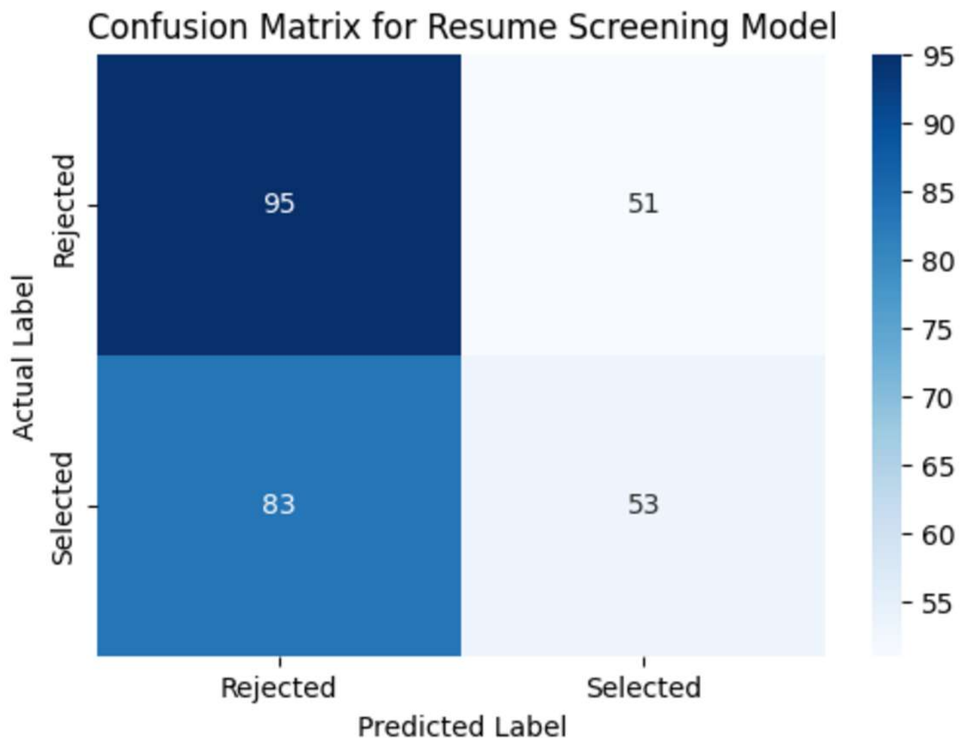
5. Confusion Matrix

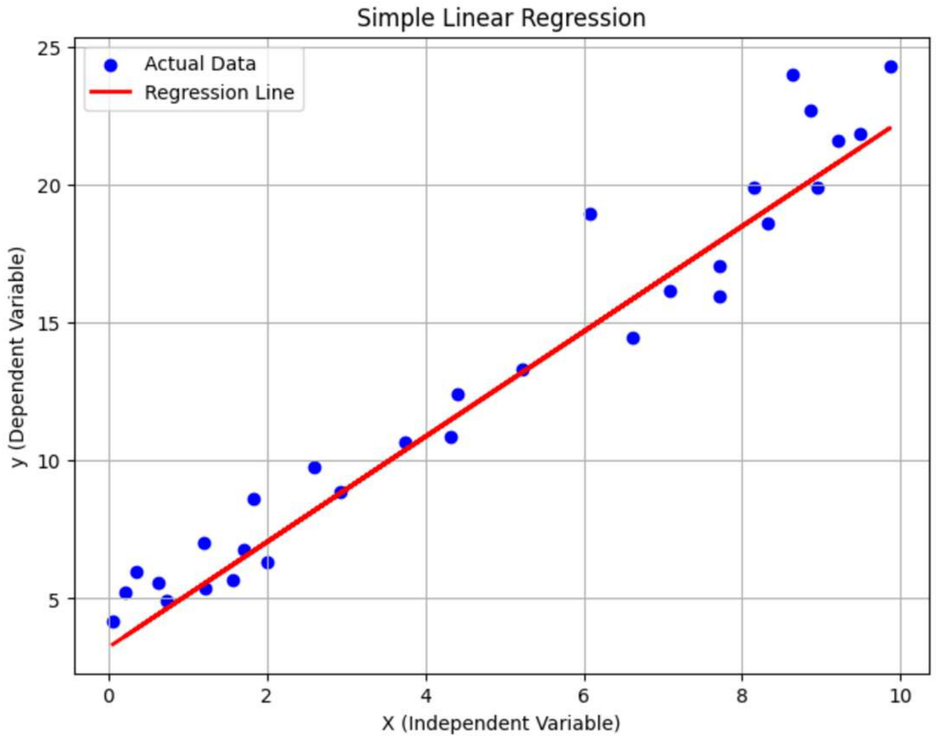
Definition:

A table that shows actual vs predicted values.

	Predicted Selected	Predicted Rejected
Actual Selected	TP	FN
Actual Rejected	FP	TN

Figures / Plots Showing Results:





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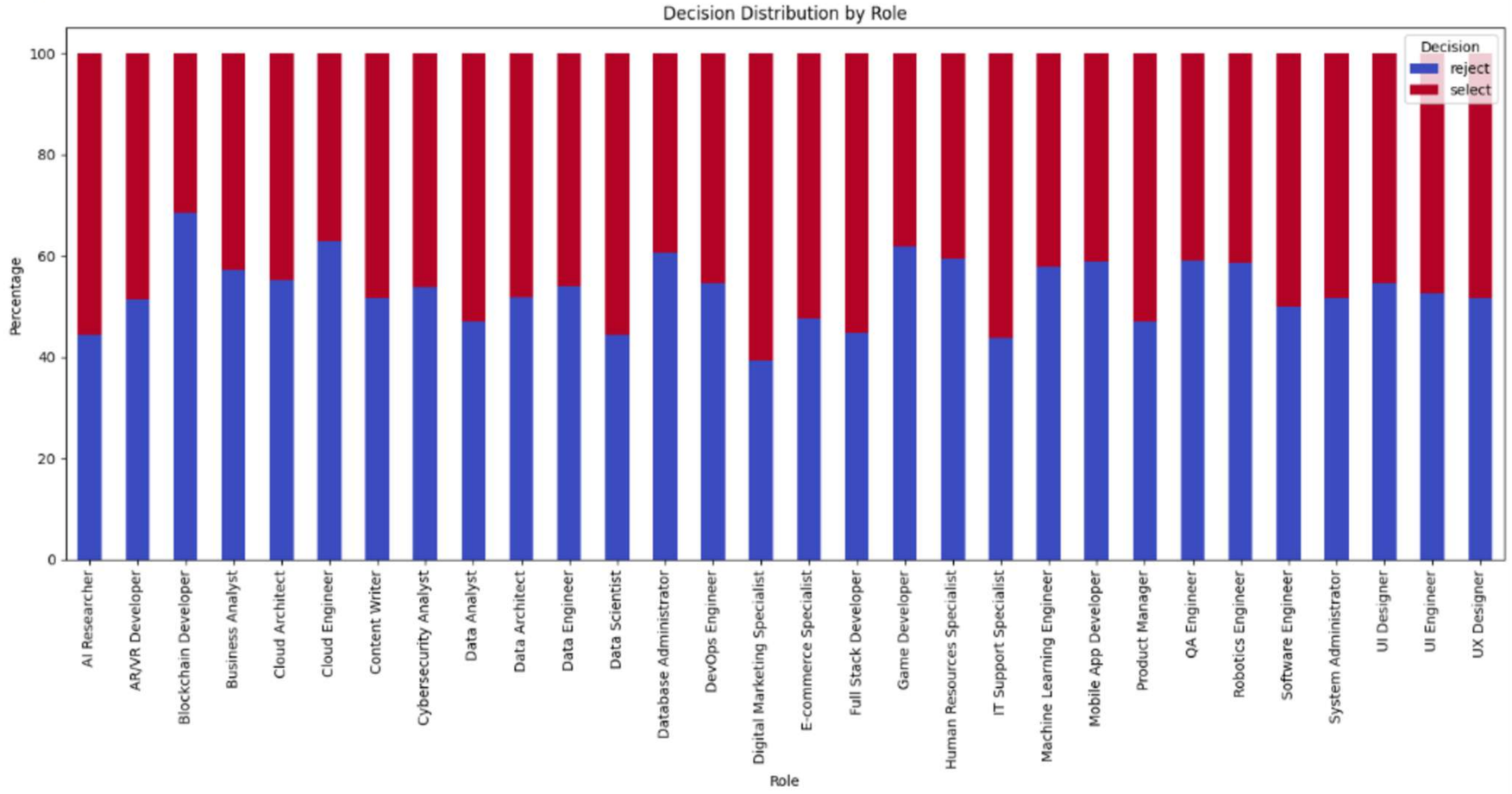
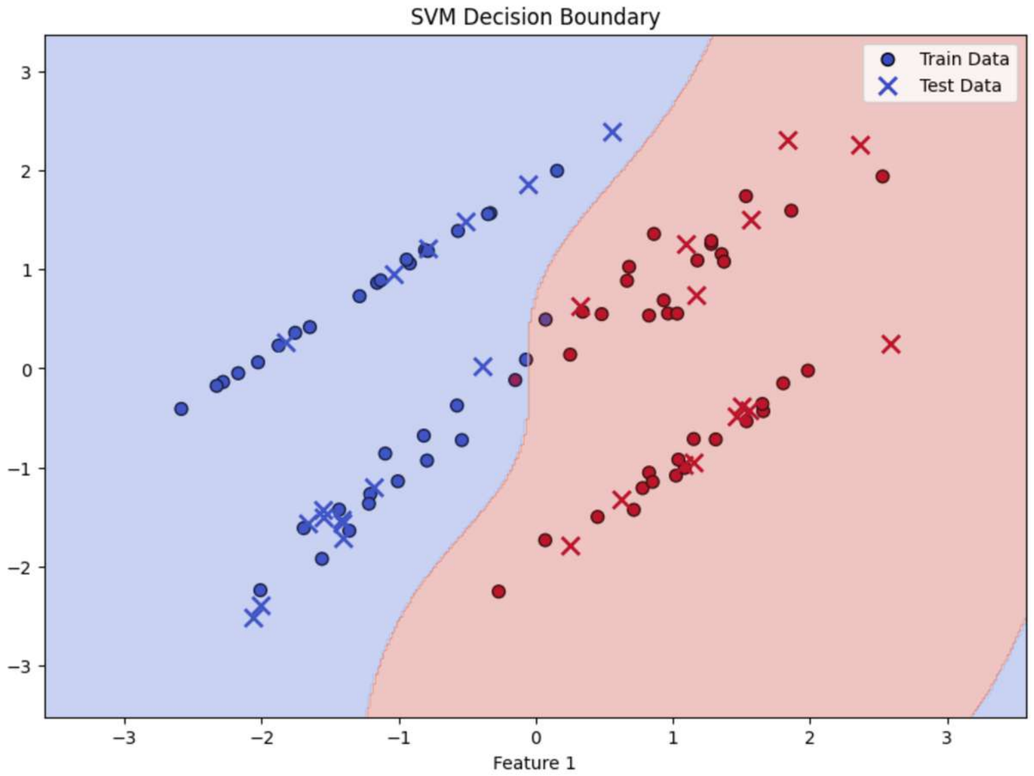
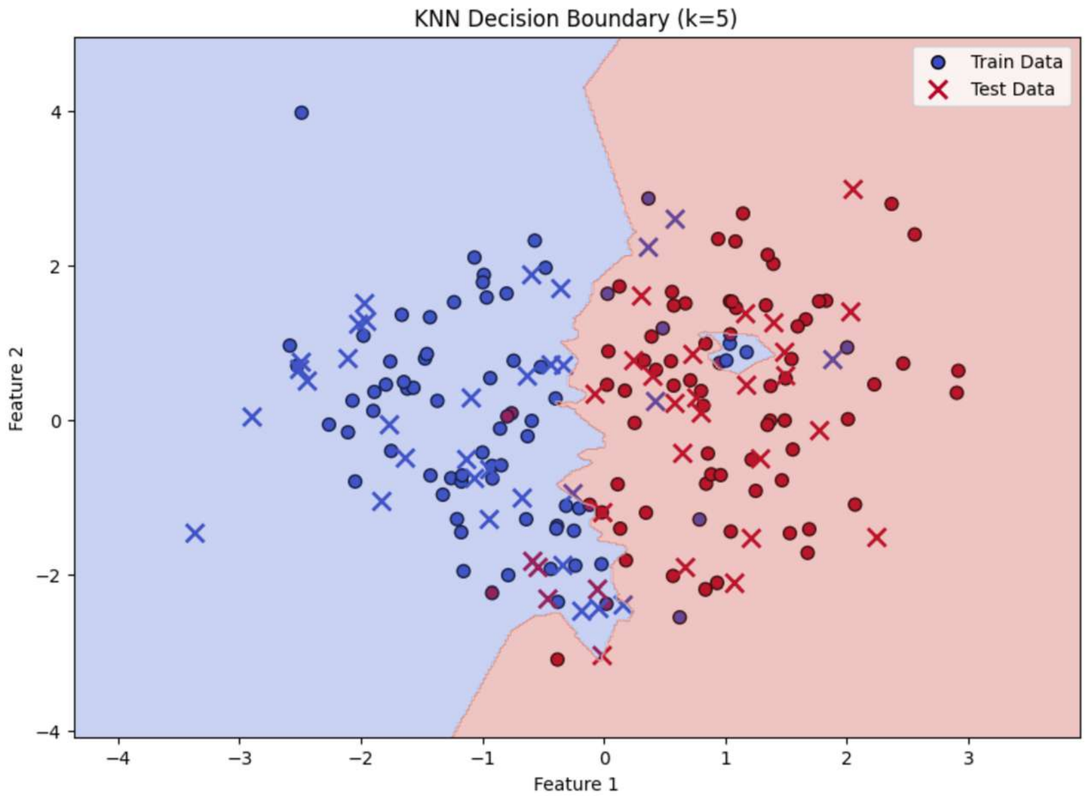
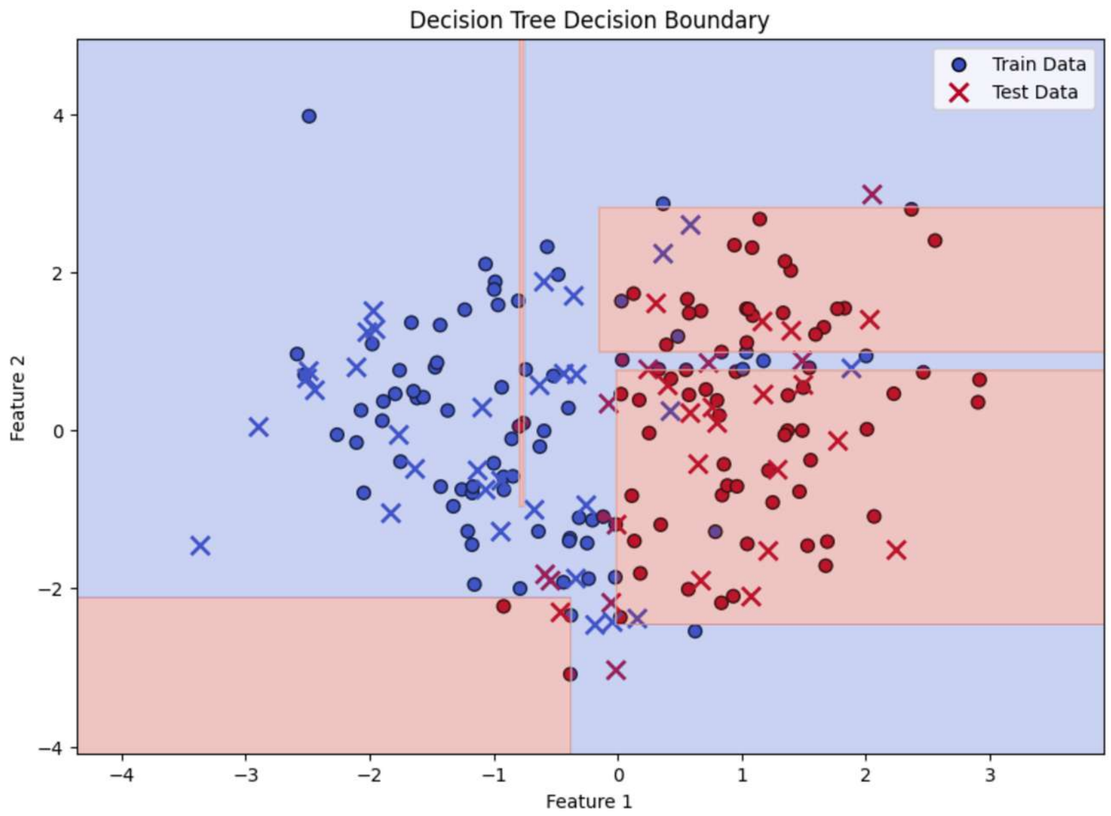
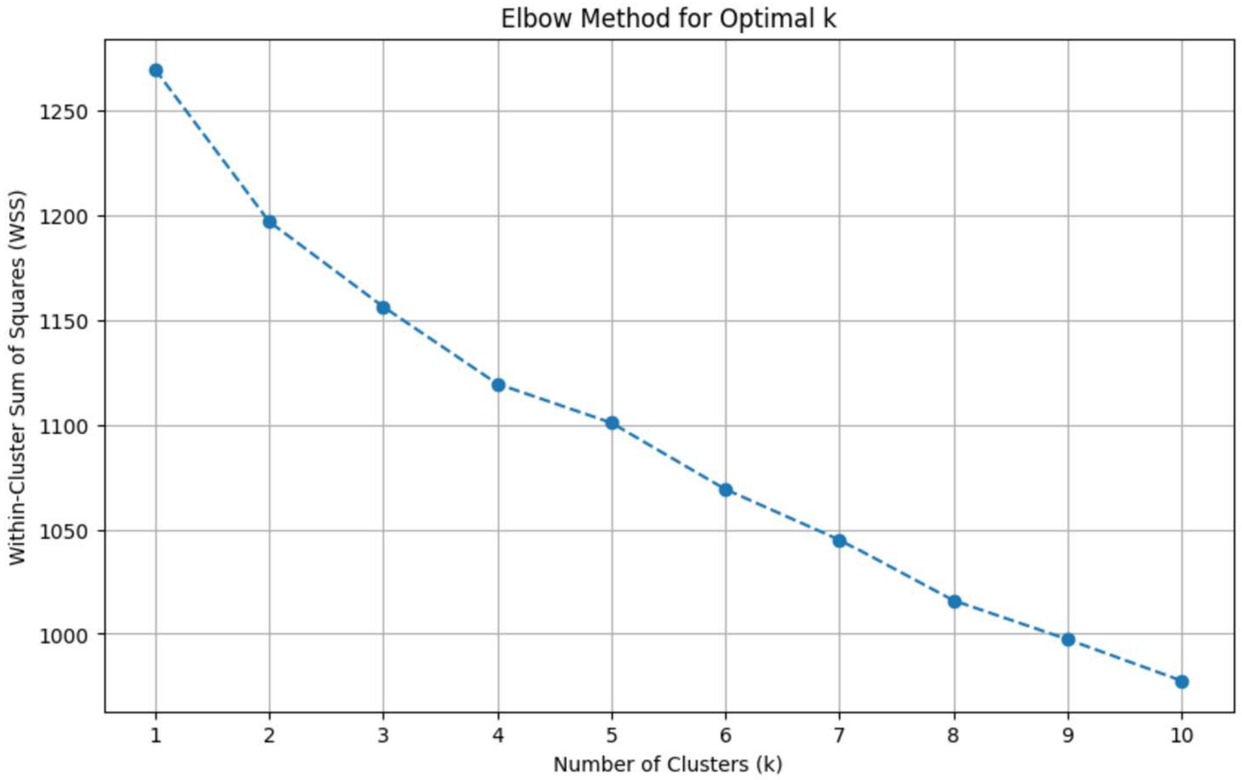



SVM Decision Boundary
Train Data
Test Data
Feature 1
3
2
1
0
−1
−2
−3


KNN Decision Boundary (k=5)
Train Data
Test Data
Feature 1
Feature 2
4
2
0
−2
−4
3


Decision Tree Decision Boundary
Train Data
Test Data
Feature 2
Feature 1
4
2
0
−2
−4
−4
−3
−2
−1
0
1
2
3


Elbow Method for Optimal k
Within-Cluster Sum of Squares (WSS)
Number of Clusters (k)
1250
1200
1150
1100
1050
1000
1
2
3
4
5
6
7
8
9
10

Conclusion

- The project successfully developed a Resume Classification System using machine learning techniques.
- The system can automatically analyze resumes and classify them based on their quality and relevance.
- Text preprocessing and feature extraction using TF-IDF improved the model's ability to understand resume content.
- The Logistic Regression model provided reliable classification results with good accuracy.
- Evaluation using accuracy, precision, recall, F1-score, and confusion matrix shows effective performance.

GitHub Repository

Project Source Code:

https://github.com/2403a51185/Resume-Classifier

https://github.com/2403a51185/Resume-Classifier/blob/main/RESUME__QUALITATIVE_DETECTOR_PROJECT.ipynb

THANK YOU